AC characterization of a metal-semiconductor diode through the lock-in technique

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June 2022

Abstract

An Al/Si metal-semiconductor Schottky diode is characterized, determining the doping type, doping density and flat band voltage. The silicon is found to be n-type through its I-V characteristics. Using a current preamplifier and a lock-in amplifier a Mott-Schottky analysis ($1/C^2$ vs. offset V) is conducted. The flat band voltage V_{bi} is estimated as $(-0.66 \pm 0.04)\,\mathrm{V}$ and the dopant density N_D as $(2.87 \pm 0.04) \times 10^{14}\,\mathrm{cm}^{-3}$.

Introduction

A metal–semiconductor (M–S) junction is a type of junction that can be used to create a rectifying Schottky diode.

M-S junctions at equilibrium display a Schottky potential barrier due to band bending, as shown in Fig.(1). The height of the barrier depends on the difference between the work functions of the metal and the semiconductor. This barrier creates a depletion region, where a lot less carriers are present, on the semiconductor side of the junction.

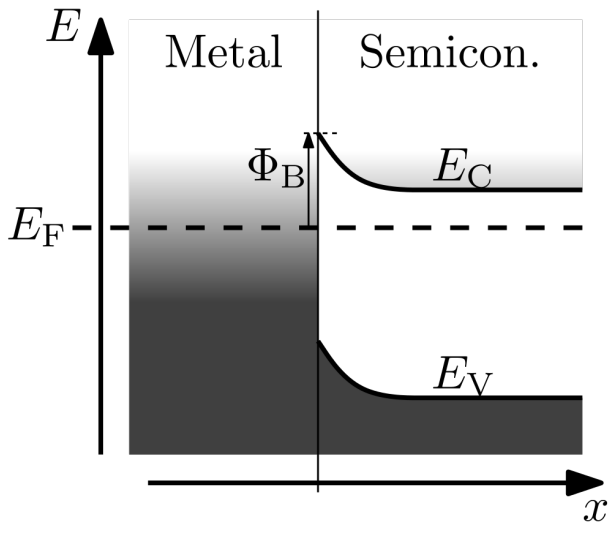


Figure 1: Band diagram of a M-S junction with an n-doped semiconductor

The size of the depletion region (and thus the diode conductivity) can be controlled by forward or reverse biasing the diode, as the voltage applied to the junction will shift the Fermi level decreasing or increasing the barrier height.

For an n-type semiconductor the depletion region thickness w can be found by solving Poisson's equations with the full depletion approximation (also called abrupt approximation), in which the charge density is assumed to be box-like and given only by the doping ions density N_D:

$$w = \sqrt{\frac{2\epsilon}{eN_D}\,(V - V_{bi})}$$

where ϵ is the dielectric constant of the semiconductor and V_{bi} is the built-in (flat band) potential that is defined as the potential at which $w = 0$ (the bands are flat and thus no depletion layer).

The Mott–Schottky equation that relates the capacitance of the junction C of area A to the applied voltage V is:

$$1/C^2 = \left(\frac{w}{\epsilon A}\right)^2 = \left(\frac{1}{\epsilon A}\right)^2 \left[\frac{2\epsilon}{eN_D}(V - V_{bi})\right]$$

Introducing a Boltzmann smearing on the charge distribution in the depletion region, a correction term will appear:

$$1/C^2 = \left(\frac{1}{\epsilon A}\right)^2 \left[\frac{2\epsilon}{eN_D}(V - V_{bi} - V_T)\right] \qquad (1)$$

where $V_T = k_B T/e$ is the thermal voltage. At 300 K $V_T \approx 26\,\text{mV}$.

The expected I-V characteristics for a Schottky diodes is the standard diode's:

$$I(V) = I_0 \left(e^{V/\eta V_T} - 1\right)$$

where η is the ideality factor and I_0 is the reverse bias saturation current.

Materials and methods

The Schottky diode under test is realized by evaporating a layer of aluminum onto a piece of doped silicon. The M-S junctions area is given as $A = 1.7 \times 1.7\,\text{mm}^2$.

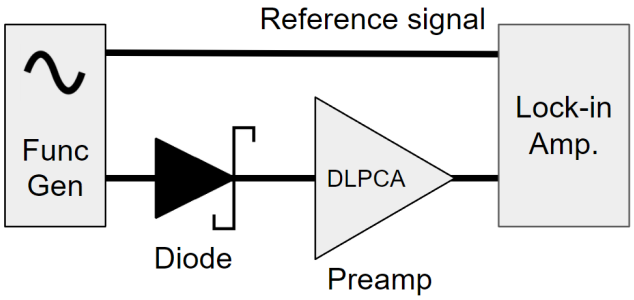


Figure 2: Schematics of the setup

The diode is kept isolated inside a metal box with BNC connectors on its sides. To probe diode's characteristics, as shown in Fig.(2), a Keysight Function Generator 33500B [1] is used to produce an AC signal with DC offset so that a lock-in amplifier can be used to probe the diode's capacitance. The current signal is preamplified and converted to voltage through a Femto DLPCA-200 current amplifier [2] before finally reaching the Stanford SR830 Lock-In Amplifier [3]. The current preamplifier is kept in the L (Low noise) FBW (full bandwidth) GND (no offset) configuration.

First, without using the lock-in, a DC analysis of the diode is conducted to obtain its I-V characteristics by using the function generator as a DC source and a digital multimeter to measure the output reported by the preamp. The gain of the preamplifier can be varied up to six orders of magnitude, accommodating easily the whole current range from reverse to forward bias.

Secondly, once the phase reading of the lock-in is calibrated, to get familiar with the workings of the lock-in technique, the measurement of a small ceramic capacitor (1 nF) is carried out. The function generator is set to create a sinusoidal signal of 10 mVrms amplitude with an offset of 0.5 V. The lock-in amplifier is connected to the preamplifier output and reference signal is fed directly to the lock-in from the function generator as shown in Fig.(2), except that instead of the diode, the capacitor is placed. To measure this capacitance a particular frequency is identified: the phase shift θ between the input and the reference signal measured by the lock-in must be 90°. The capacitance is then given by:

$$C = \frac{V_{out}}{G V_{in}} \frac{1}{2\pi f} \tag{2}$$

where G is the gain of the preamplifier and f is the frequency. At $f = 800\,\text{Hz}$ the test capacitance is estimated at $0.930 \pm 0.009\,\text{nF}$, to be compared to the nominal value of 1 nF.

Finally to obtain the Mott-Schottky plot of the diode, the function generator signal AC amplitude is set to 10 mVrms, DC offset is varied from 4.5 V down to -0.2 V while the signal frequency is set at 811 Hz to avoid power grid harmonics. The gain G of the preamp is kept at 10^6 V/A while the sensitivity of the lock-in is varied between different values to avoid overload. Reserve is set to "Normal", time constant and filter order are chosen to have a decent refresh rate of the impedance; exact values and other parameters are tabulated below in Tab.(1).

Lock-In Amplifier parameters	
Coupling	AC
CH1	R
CH2	θ
Sensitivity	5-200 mV
Time constant	100 ms
Filter order	3
Reserve	Normal
Notch	1x, 2x

Table 1: Lock-in parameter

By modelling the diode in reverse biasing as a RC parallel circuit each capacitance data point obtained with Eq.(2) has to be corrected by a factor of $\sin(\theta)$. The flat band potential and the dopant density can be inferred directly from a linear fit of the Mott-Schottky $1/C^2$ vs. V plot (Eq.(1)): the former is given by the intercept of the fit line with the voltage axis, the latter is instead inversely proportional to the slope of the fit.

Error analysis

The error on C (Eq.(2)) is just given by the gain accuracy which is specified as 1% [2] since all other error sources are negligible. This gives a relative error on $1/C^2$ of 1.42%.

Instrumentation noise

The instruments are a source of known and quantifiable white noise and thus are subtracted from the output RMS voltage read-out. It follows a list of all the noise sources considered which are then added under quadrature as they are assumed uncorrelated.

At the preamplifier's input: the current noise due to input noise voltage density ($4\ \mathrm{nV}/\sqrt{\mathrm{Hz}}$) divided by the measured resistance ($>10\,\mathrm{k\Omega}$), the shot noise current density ($\sqrt{2eI}$) and input noise current density ($1.3 \times 10^{-13}\ \mathrm{A}/\sqrt{\mathrm{Hz}}$). This last contribution is the dominant one.

At the lock-in's input: the previous total noise density is multiplied by $G = 10^6\,\mathrm{V/A}$ and is added to the lock-in voltage noise ($6\ \mathrm{nV}/\sqrt{\mathrm{Hz}}$).

The equivalent noise bandwidth of a third order RC cascade filter is $0.094/\tau$ [4], where τ is the time constant (100 ms): 0.94 Hz.

The total noise removed is merely 0.1 mV and so it is actually negligible.

Results and discussion

The I-V characteristic is displayed in Fig.(3). Voltage is referred to the metal surface. From this characteristic it can be deduced that the doping of the Si is of n-type since the electrons have the energy to cross the barrier only when a negative potential is applied to the silicon.

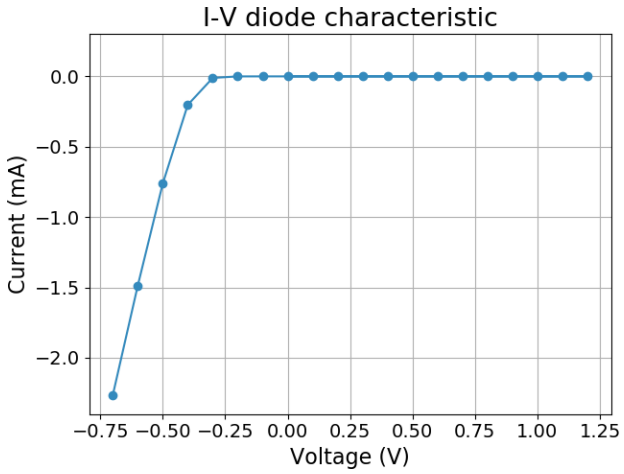


Figure 3: I-V characteristics of the diode. Voltage is applied to Silicon

Fig.(4) displays the Mott-Schottky plot of the diode. The fit is conducted in the reverse region, from $V = 0.2\,\mathrm{V}$ onwards, where the phase remains approximately 90°. The fit has an R^2 of 0.998 and its intercept directly estimates the flat band potential at $V_{bi} = -0.66 \pm 0.04\,\mathrm{V}$. The fit already takes into account the Boltzmann correction using $V_T = 26\,\mathrm{mV}$. The dopant density N_D, using a relative permittivity of Si of 11.7 [5], is estimated as $(2.87 \pm 0.04) \times 10^{14}\,\mathrm{cm}^{-3}$.

Both of these values are reasonable for the type of diode examined.

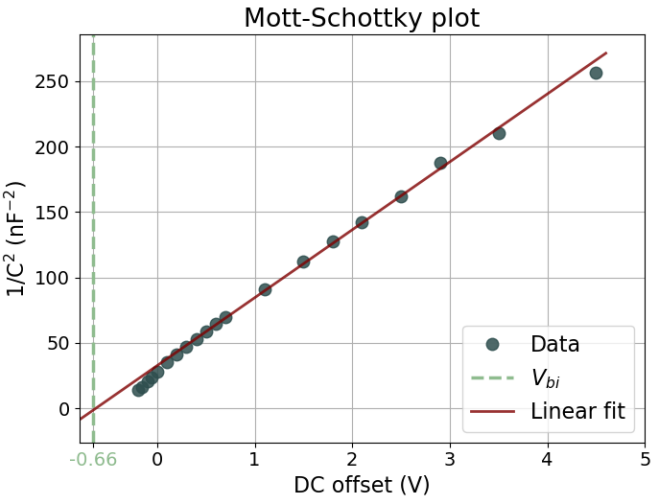


Figure 4: Mott-Schottky plot of the diode. The flat band potential V_{bi} is estimated as 0.66 V, considering Boltzmann correction

Conclusions

Thanks to the lock-in technique it is possible to measure small signals and impedances. This technique is used in this work to perform an AC characterization of a Schottky diode composed of Al and n-doped Si.

The doping type of the Si is confirmed through and I-V plot, then to get familiar with the lock-in amplifier a small capacitance is measured and the uncertainty on this measure is estimated.

From a linear fit of the Mott-Schottky plot of the diode the flat band voltage is estimated to be $V_{bi} = -0.66 \pm 0.04\,\mathrm{V}$, while the dopant density N_D is calculated to be $(2.87 \pm 0.04) \times 10^{14}\,\mathrm{cm}^{-3}$.

References

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[5] W. C. Dunlap and R. L. Watters. "Direct Measurement of the Dielectric Constants of Silicon and Germanium". In: *Phys. Rev.* 92 (6 Dec. 1953), pp. 1396–1397. DOI: 10.1103/PhysRev.92.1396. URL: https://link.aps.org/doi/10.1103/PhysRev.92.1396.